

Functional Test Suite Report: Login Authentication Module

1. Project Overview & Request Context

The objective of this document is to define and formalize a functional test suite for a standardized Login Authentication Module. This report ensures that the module meets all technical requirements for security, data integrity, and user experience by validating input constraints, authentication logic, and navigational flow. The following requirements served as the foundation for this test suite: Generate 8 test cases for a simple login page with username (email), password (min 8 chars), login button, and forgot password link. Include 2 negative and 1 boundary value test case.

2. System Under Test (SUT) Specifications

The testing scope encompasses the following UI components and functional constraints as derived from the requirements:

- **Input Fields:**
- **Username/Email:** Requires standard syntax validation, specifically the inclusion of the "@" character.
- **Password:** Subject to a minimum length constraint of 8 characters and strict case-sensitivity.
- **Functional Requirements:**
- The system must enforce a minimum password length of 8 characters.
- The system must support and process character inputs up to a 100-character boundary.
- **Interactive Elements:**
- **Login Button:** Executes the authentication logic and session initiation.
- **Forgot Password Link:** Provides a navigational path to the account recovery module.

3. Detailed Functional Test Suite

The table below details the comprehensive test suite, including specific input data and expected technical outcomes.

Test ID	Test Case Name	Category	Test Steps/Input	Expected Result
TC01	Valid Login	Positive	Enter valid email 'user@example.com' and 10-char password 'Secure1234'	Authentication successful; user session initiated.
TC02	Invalid Email Format	Negative	Enter 'user123' (omitting the '@' character) and valid password	Inline field-level validation message appears regarding invalid email format.
TC03	Short Password	Negative	Enter valid email and a 7-character password (e.g., 'Pass123')	Authentication rejected; error message 'Password must be at least 8 characters' displayed.

|| **TC04** | Password Boundary (Lower) | Boundary Value | Enter valid email and exactly an 8-character password (e.g., '8CharPwd'). | Authentication successful; user session initiated.

|| **TC05** | Empty Fields | Negative | Leave both Email and Password fields null; click 'Login'. | Inline field-level validation messages appear for all mandatory fields.

|| **TC06** | Forgot Password Link | Navigation | Click the 'Forgot Password' hypertext link. | Successful redirect to the account recovery/password reset page.

|| **TC07** | Password Case Sensitivity | Negative | Enter valid email and password 'P@ssword123' when the actual credential is 'p@ssword123'. | Authentication rejected; error message 'Invalid credentials' displayed.

|| **TC08** | Password Boundary (Upper) | Boundary Value | Enter valid email and a password string consisting of exactly 100 characters. | Authentication successful; user session initiated. |

4. Test Case Classification Summary

This summary provides a map of test coverage across different functional areas of the authentication module:

- **Positive Functional Tests (Happy Path):** Validates the core authentication logic under standard successful conditions.
- TC01
- **Negative Tests (Error Handling):** Verifies the system's ability to handle and reject invalid input gracefully.
- TC02, TC03, TC05, TC07
- **Boundary Value Analysis (BVA):** Targets the limits of the password length requirements to ensure edge-case stability.
- TC04 (Lower limit: 8 characters)
- TC08 (Upper limit: 100 characters)
- **Navigation/UI Tests:** Validates secondary application flows and interface elements.
- TC06

5. Technical Constraints & Observations

The following technical behaviors were observed during the design of this suite and must be maintained in the production environment:

- **Strict Case Sensitivity:** The authentication logic must treat the password field as case-sensitive to ensure security integrity (TC07).
- **Buffer/Input Limits:** The system must remain stable and permit authentication when handling large character inputs up to the 100-character boundary (TC08).
- **Mandatory Field Enforcement:** The system must prevent POST requests and trigger field-level validation if mandatory fields are left empty (TC05).
- **Email Syntax Enforcement:** The username field requires formatting validation that specifically checks for the presence of standard email indicators, such as the "@" symbol (TC02).

- **Length Logic:** The system must strictly enforce the 8-character minimum, ensuring that n-1 (7 characters) results in a rejection (TC03) while the n limit (8 characters) results in a pass (TC04).